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Kuo et al.

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(54) **CIRCUIT BOARD STRUCTURE AND MANUFACTURING METHOD THEREOF**

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H05K 1/16 (2006.01)
H05K 3/36 (2006.01)
(52) **U.S. Cl.**
CPC **H05K 1/142** (2013.01); **H05K 1/165** (2013.01); **H05K 3/368** (2013.01); **H05K 2201/086** (2013.01); **H05K 2201/10265** (2013.01)

(58) **Field of Classification Search**
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See application file for complete search history.

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Primary Examiner — Timothy J Thompson

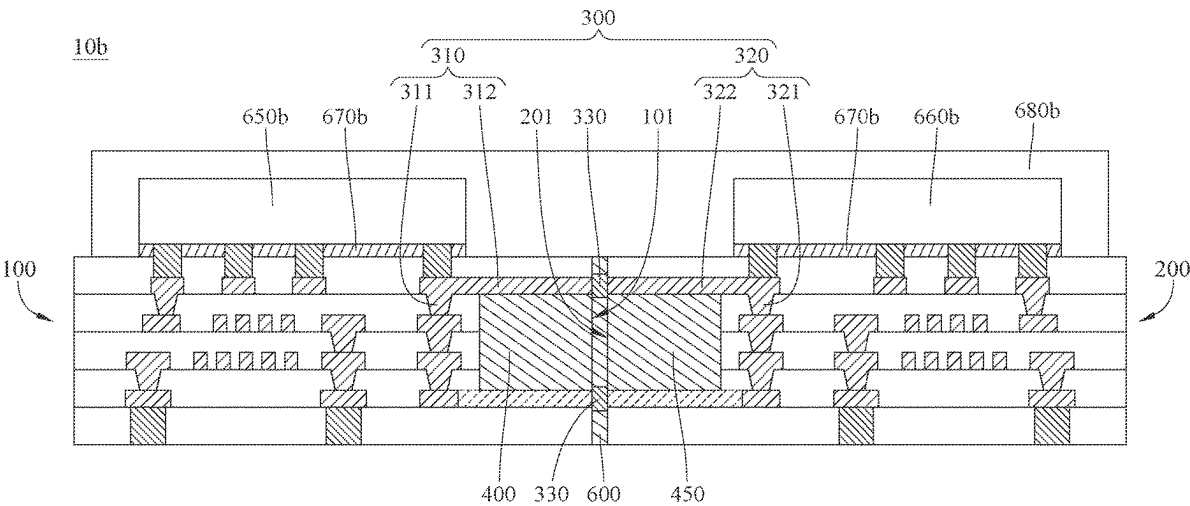
Assistant Examiner — Amol H Patel

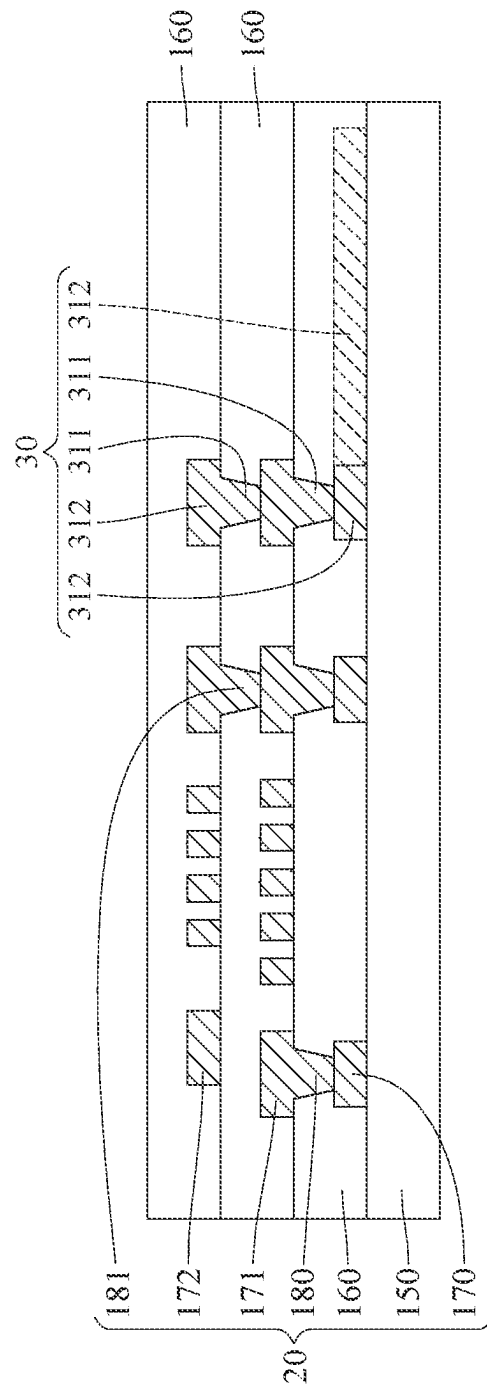
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(57) **ABSTRACT**

A circuit board structure and a manufacturing method thereof. Circuit board structure includes first circuit board, second circuit board, conductive coil, magnetic body and molding compound. First circuit board has first side surface and first cavity located on first side surface. Second circuit board has second side surface facing first side surface and being spaced apart from first side surface. Conductive coil is in a spiral shape and includes first coil pattern and second coil pattern. First coil pattern is disposed in first circuit board. Second coil pattern is disposed in second circuit board. First coil pattern is electrically connected to second coil pattern. Magnetic body is filled in first cavity of first circuit board. Conductive coil surrounds at least a part of magnetic body. Molding compound is filled in a gap between first side surface and second side surface.

10 Claims, 20 Drawing Sheets





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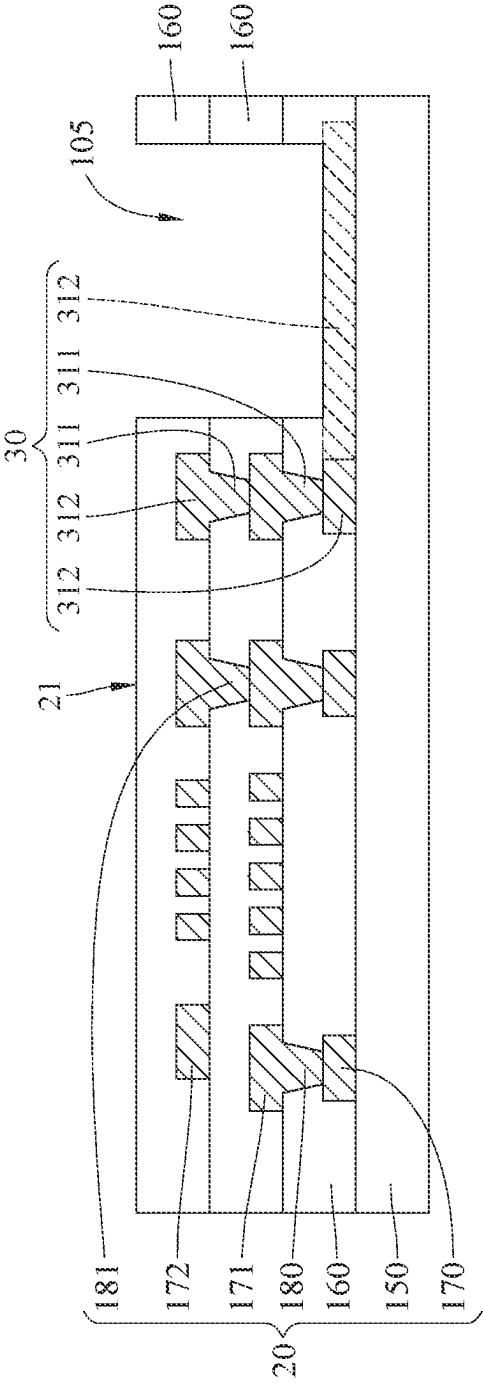


FIG. 2

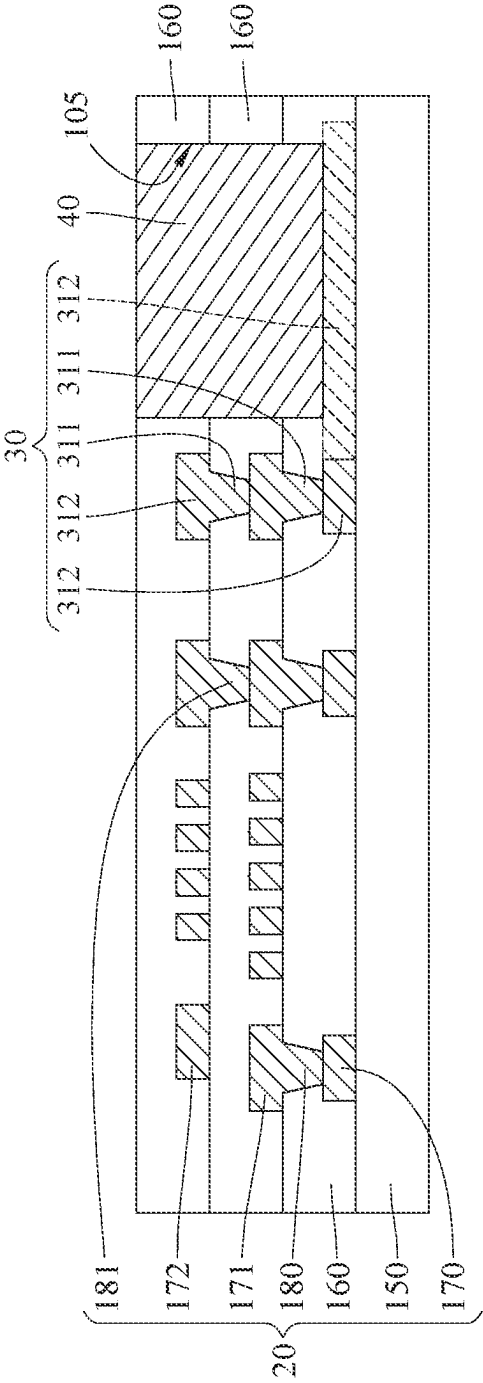


FIG. 3

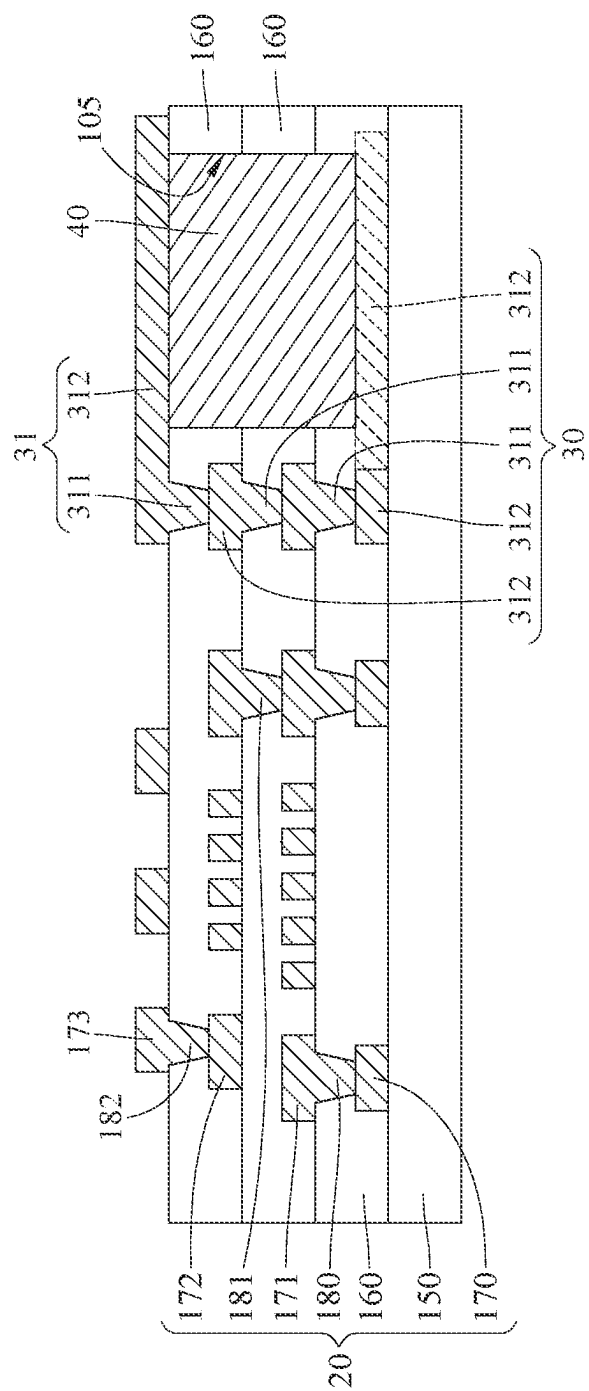
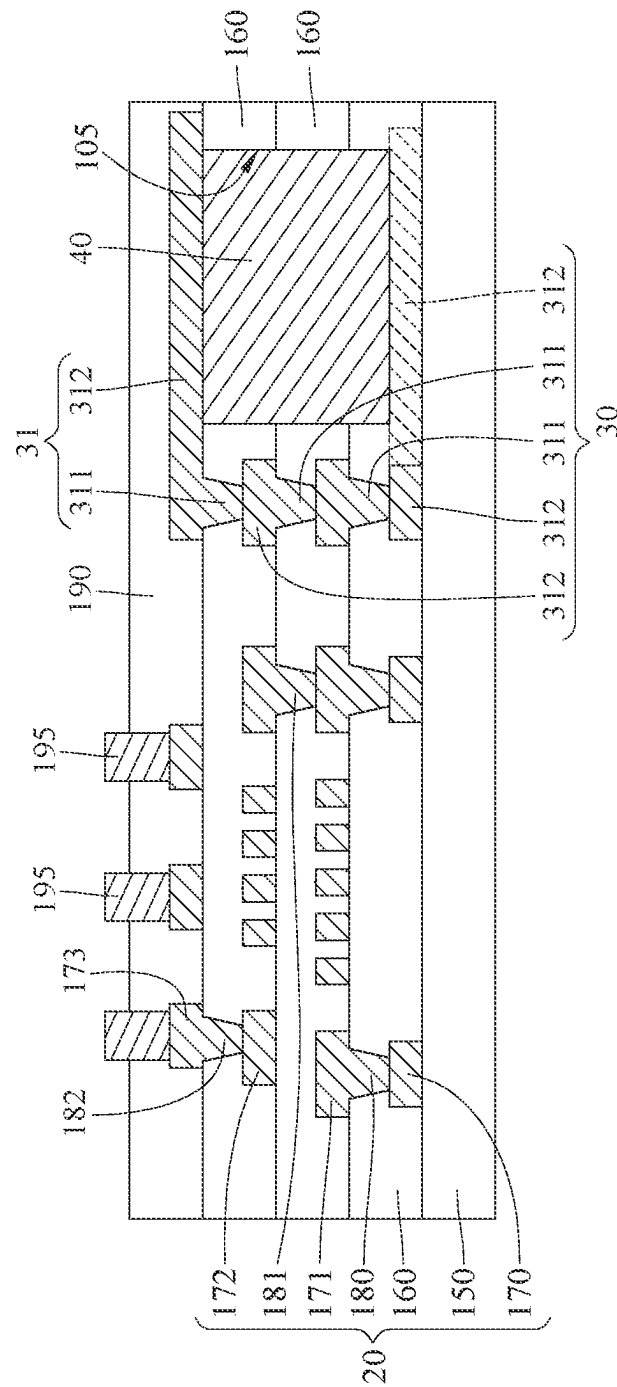
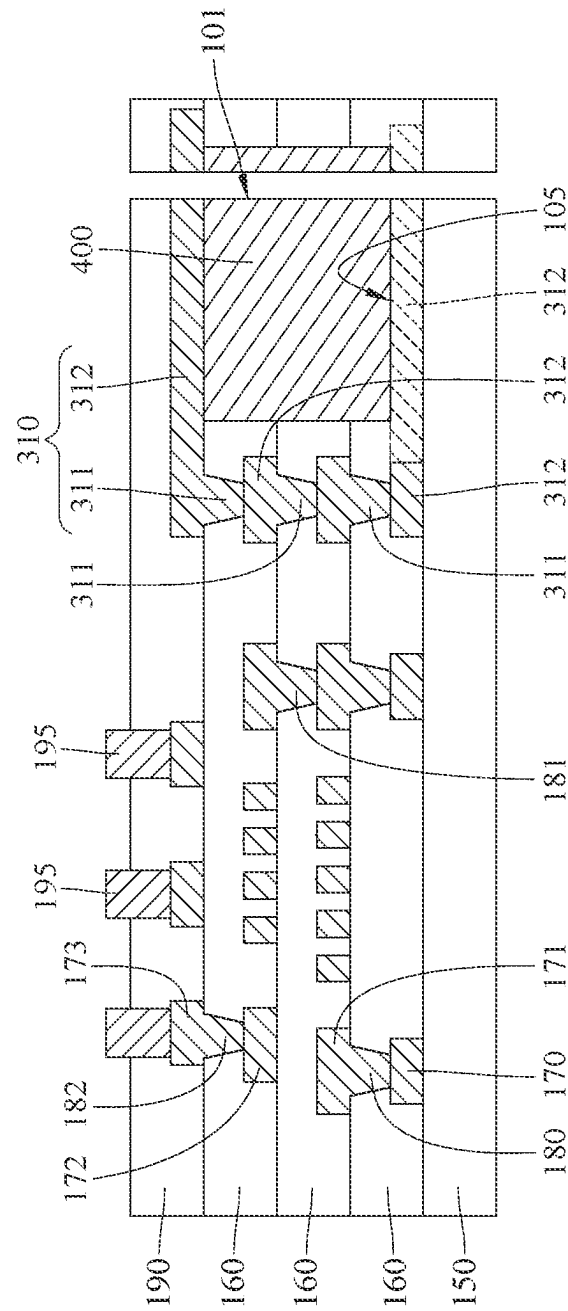


FIG. 4



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G
L

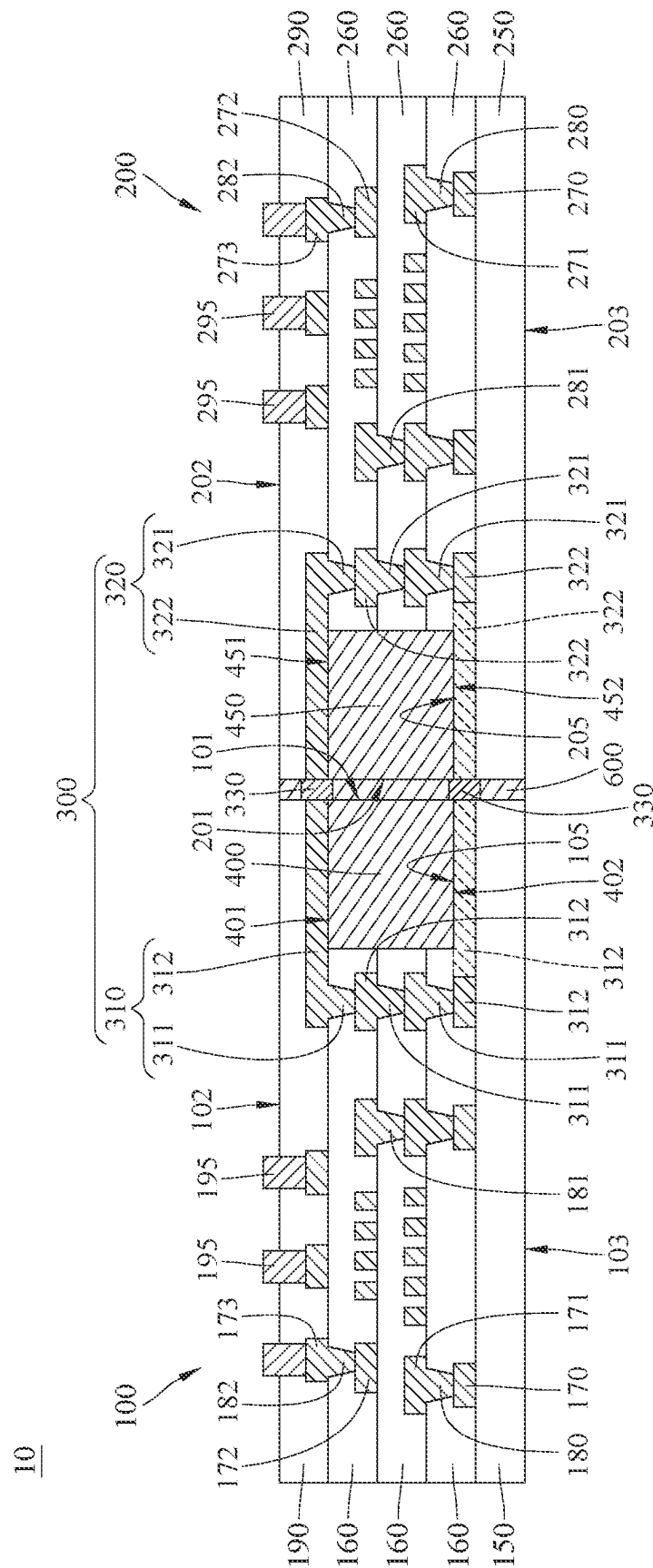


Fig. 7.

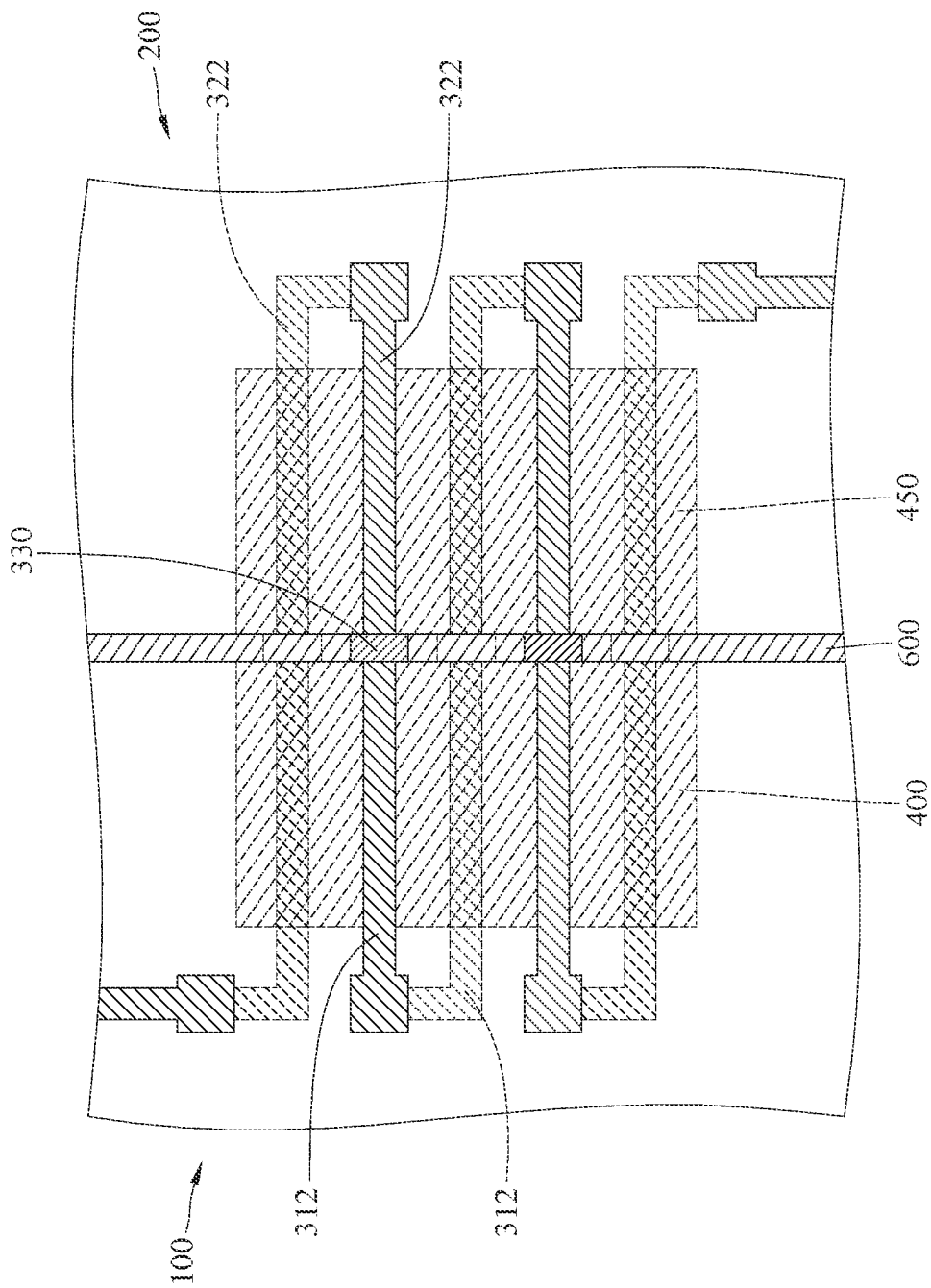


FIG. 8

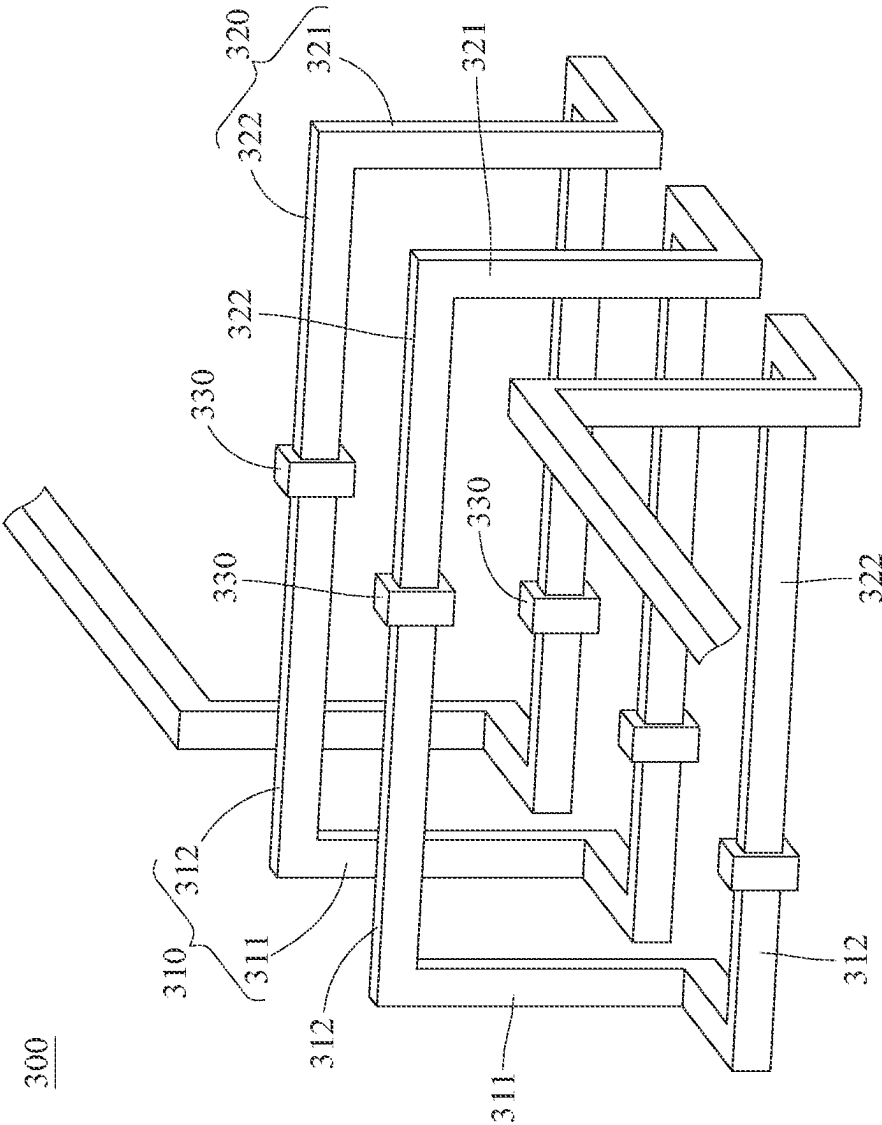
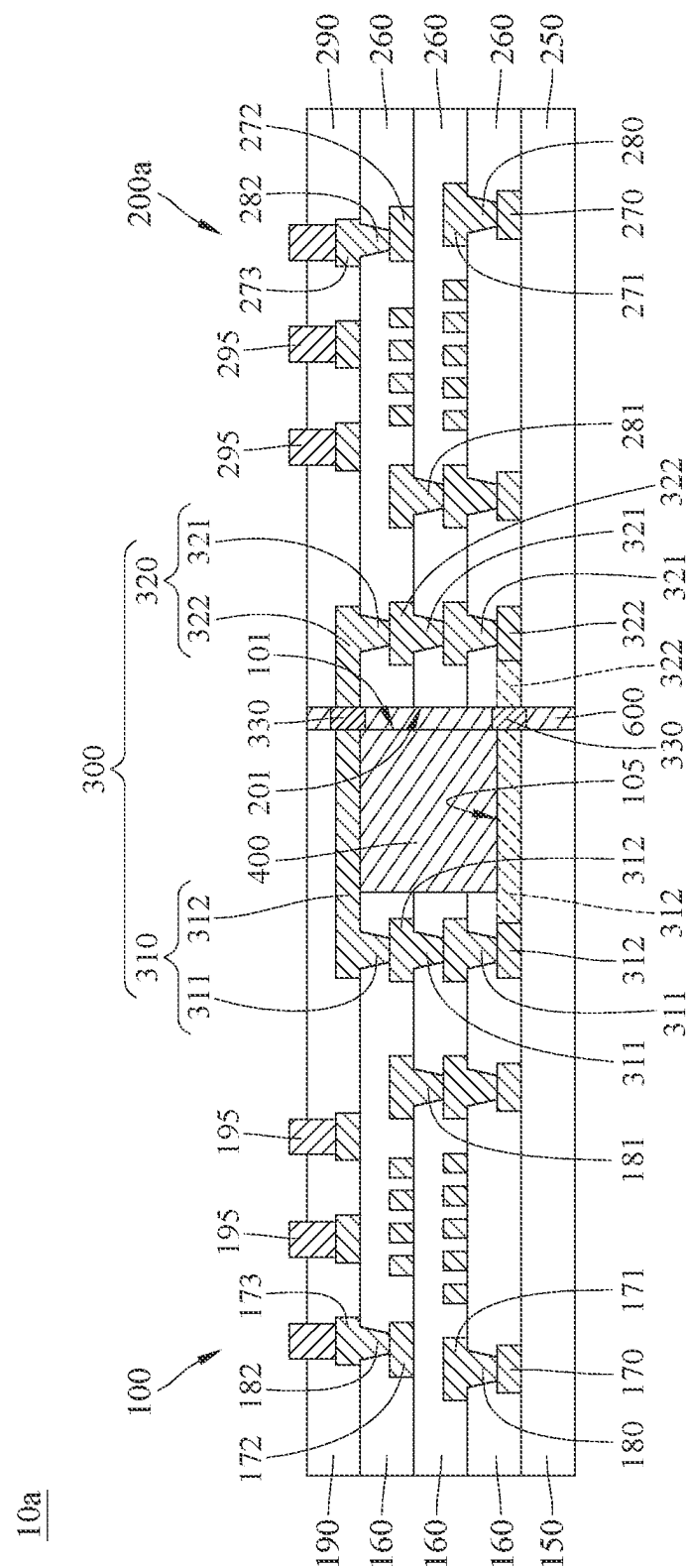


FIG. 9





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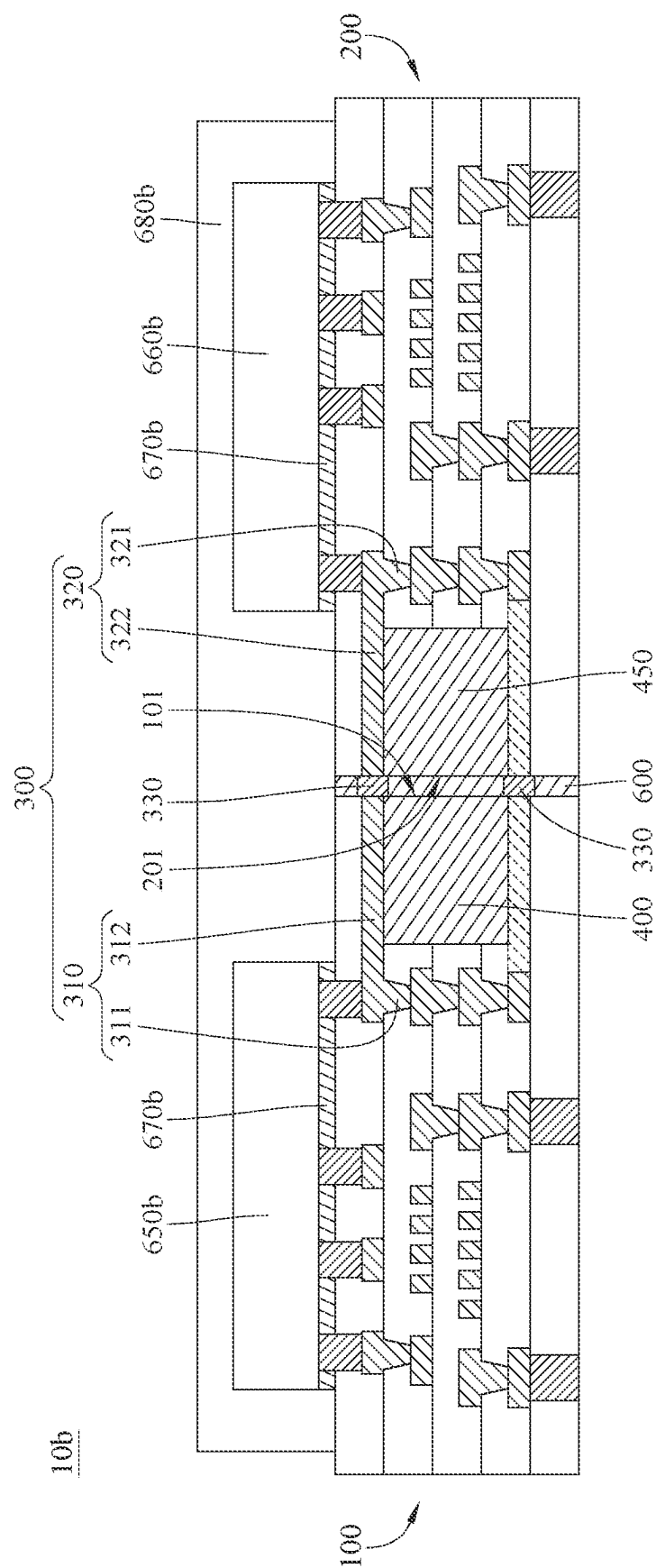



FIG. 11

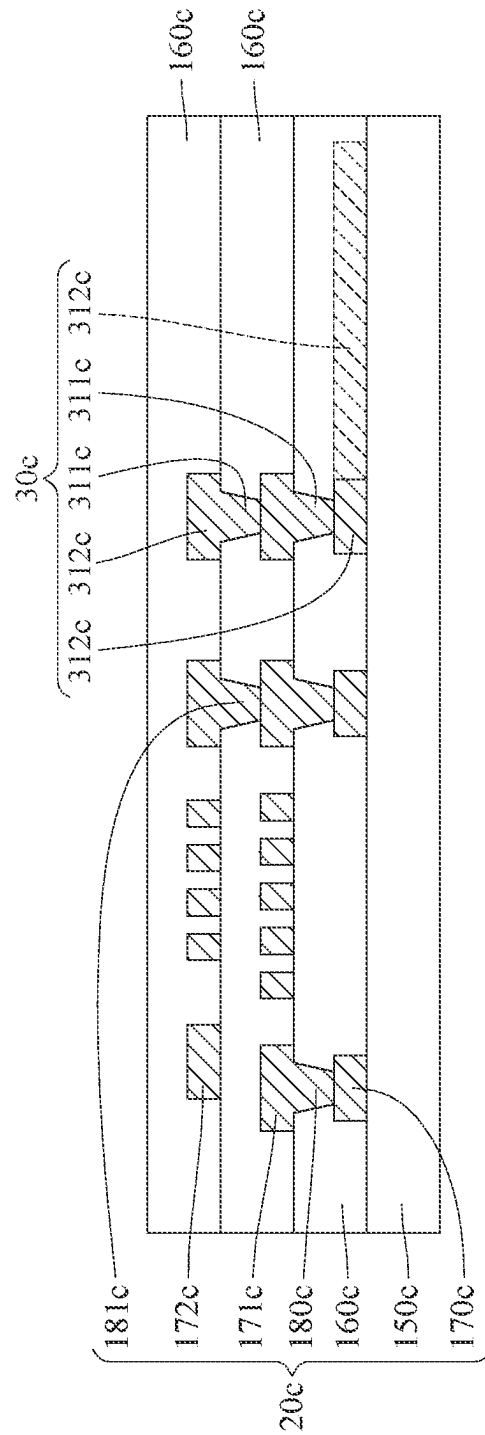
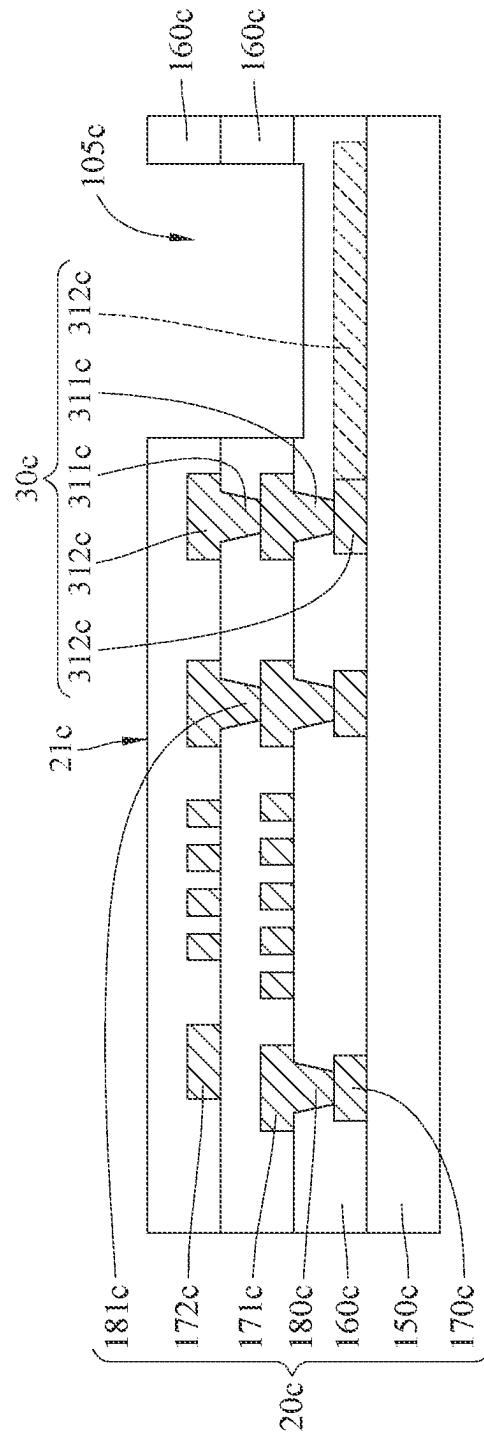
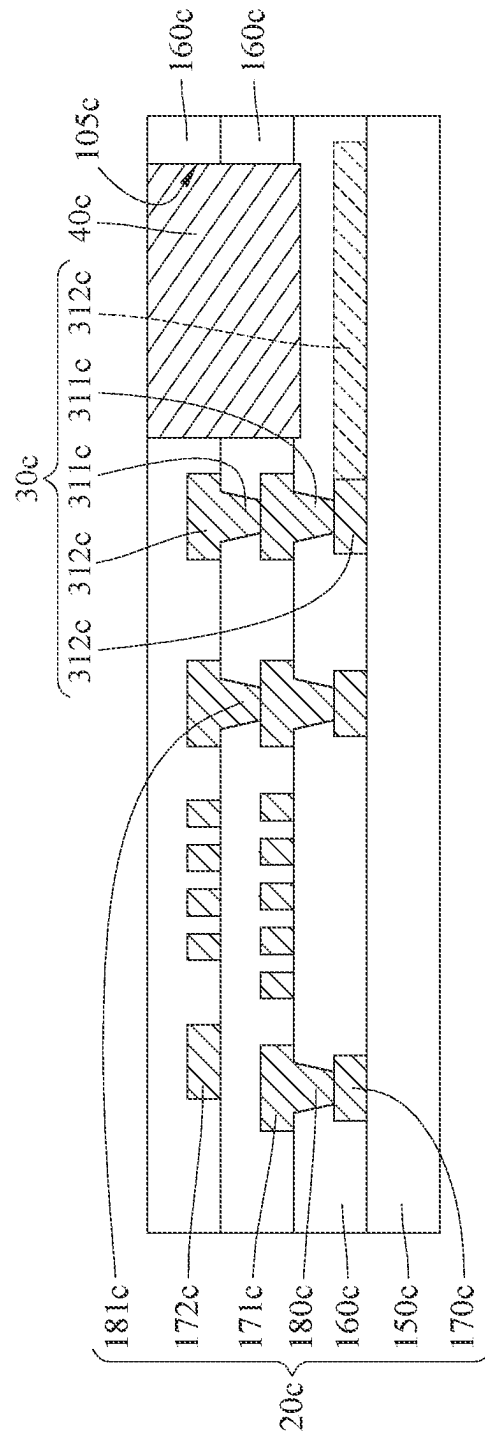


FIG. 12*

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FIG



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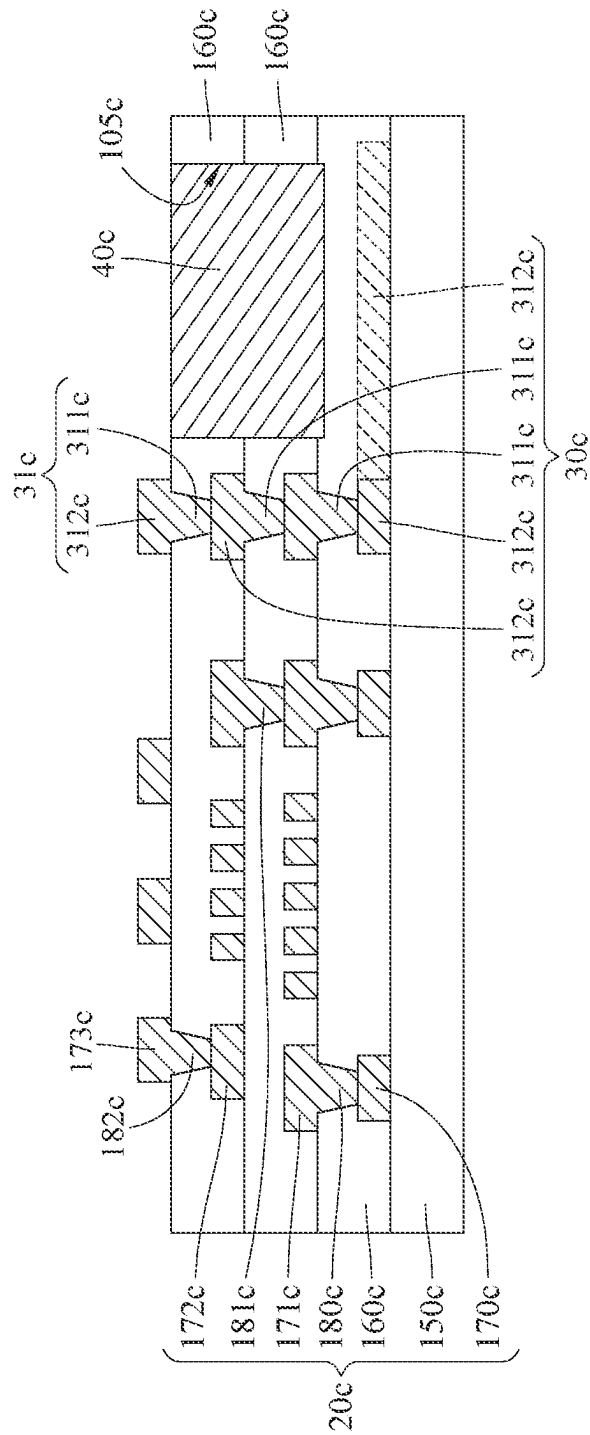


FIG. 15*

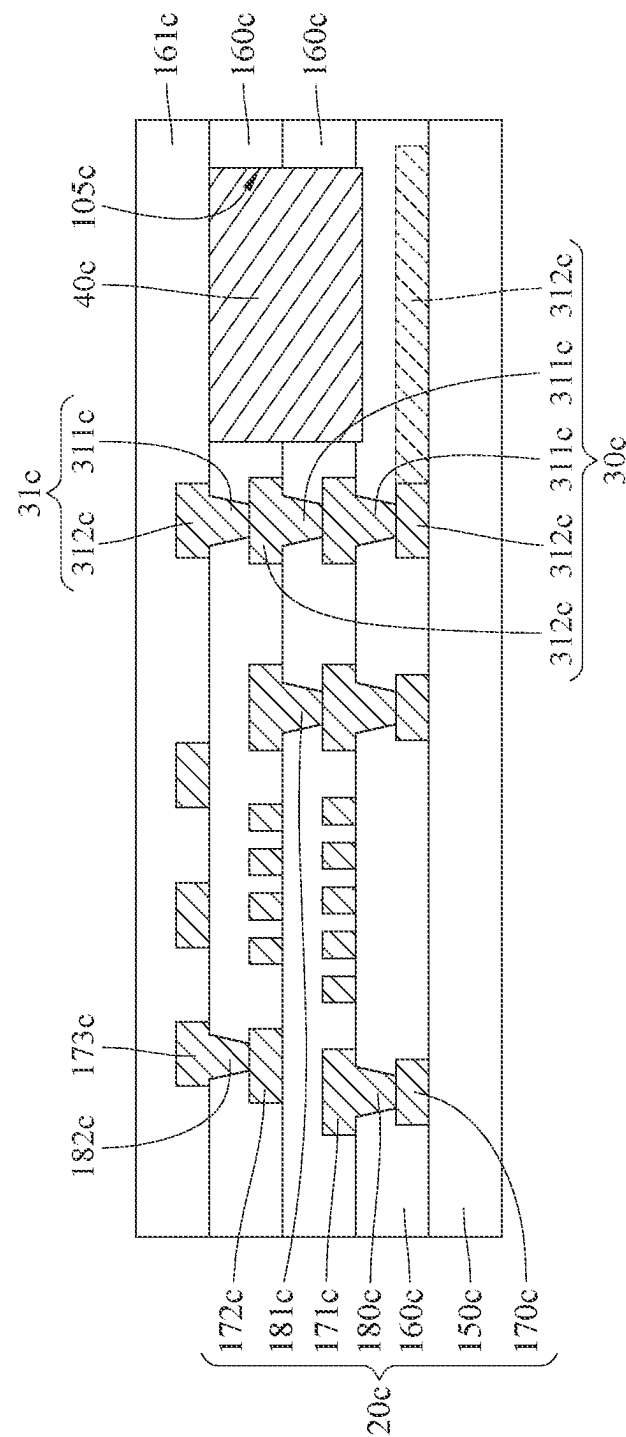


FIG. 16

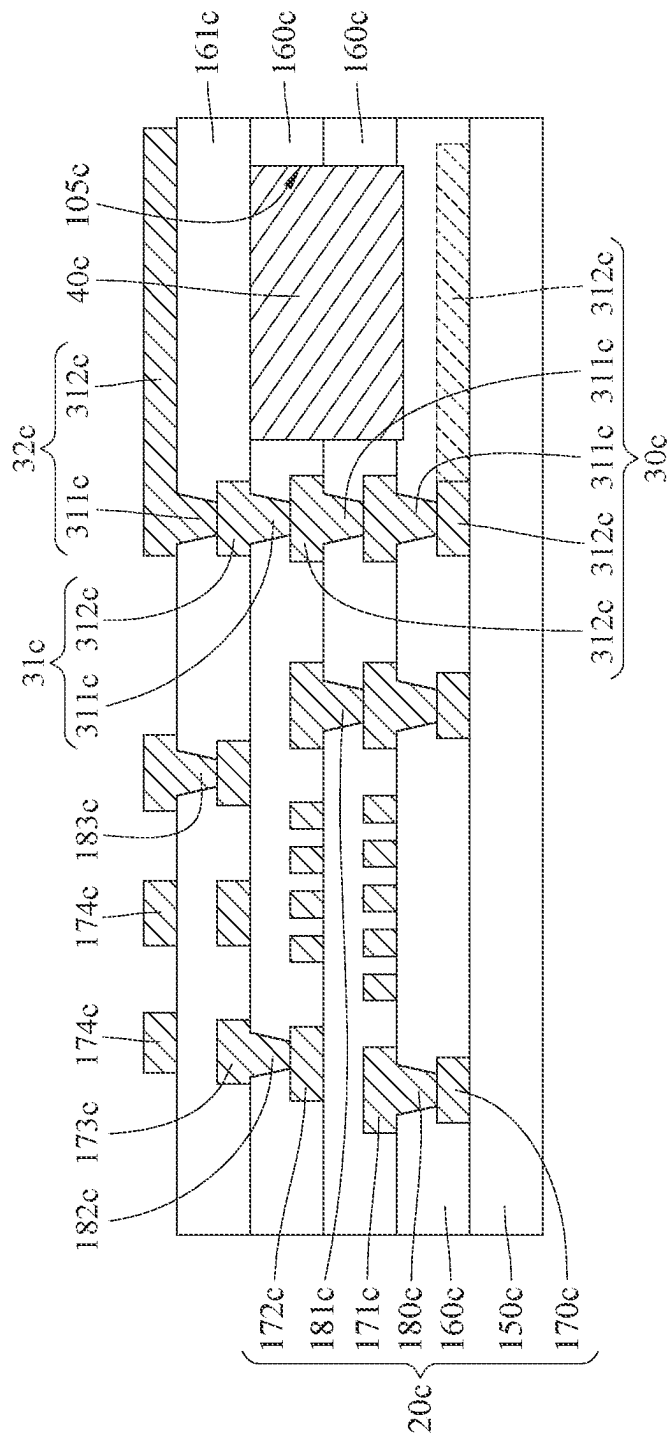


FIG. 17

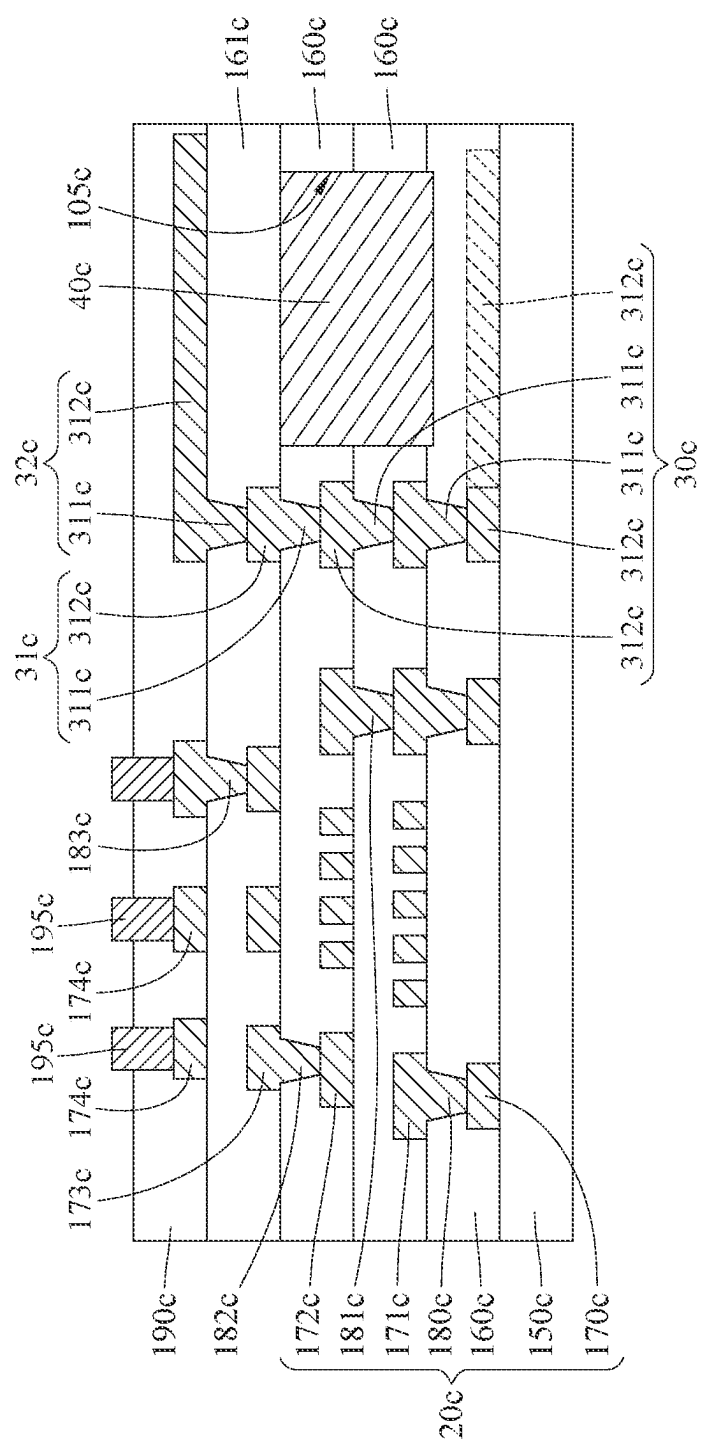
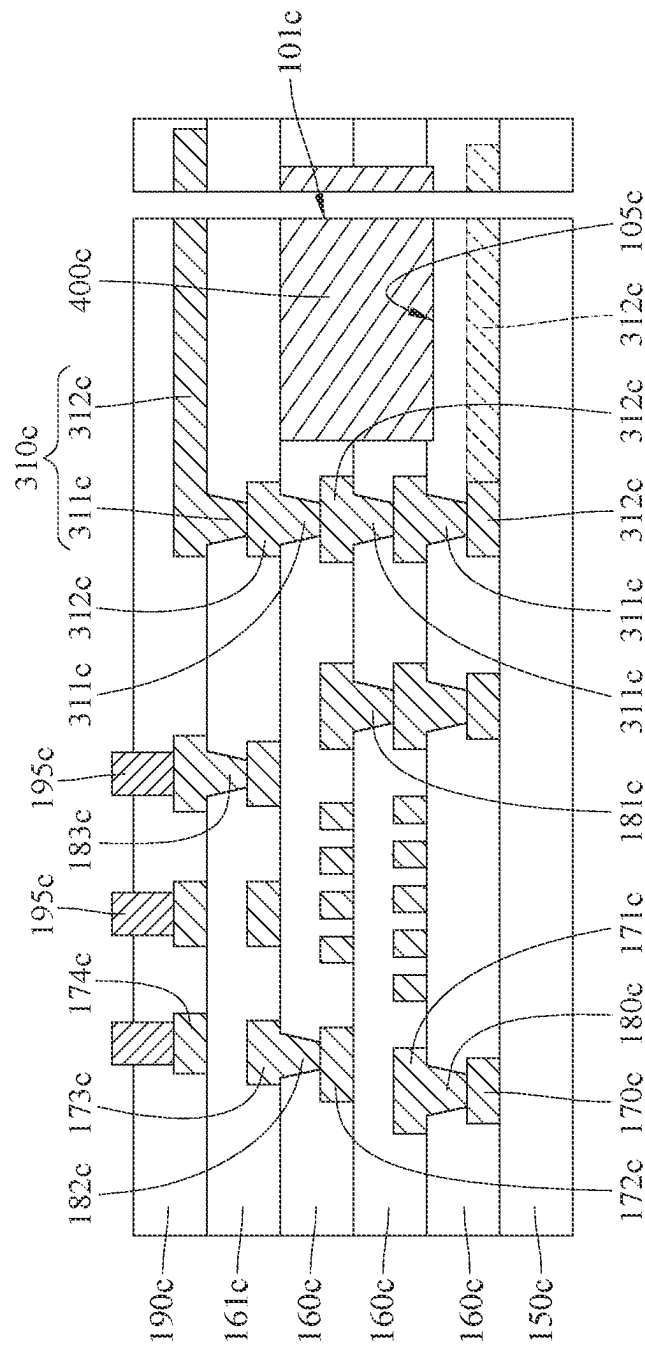


FIG. 18



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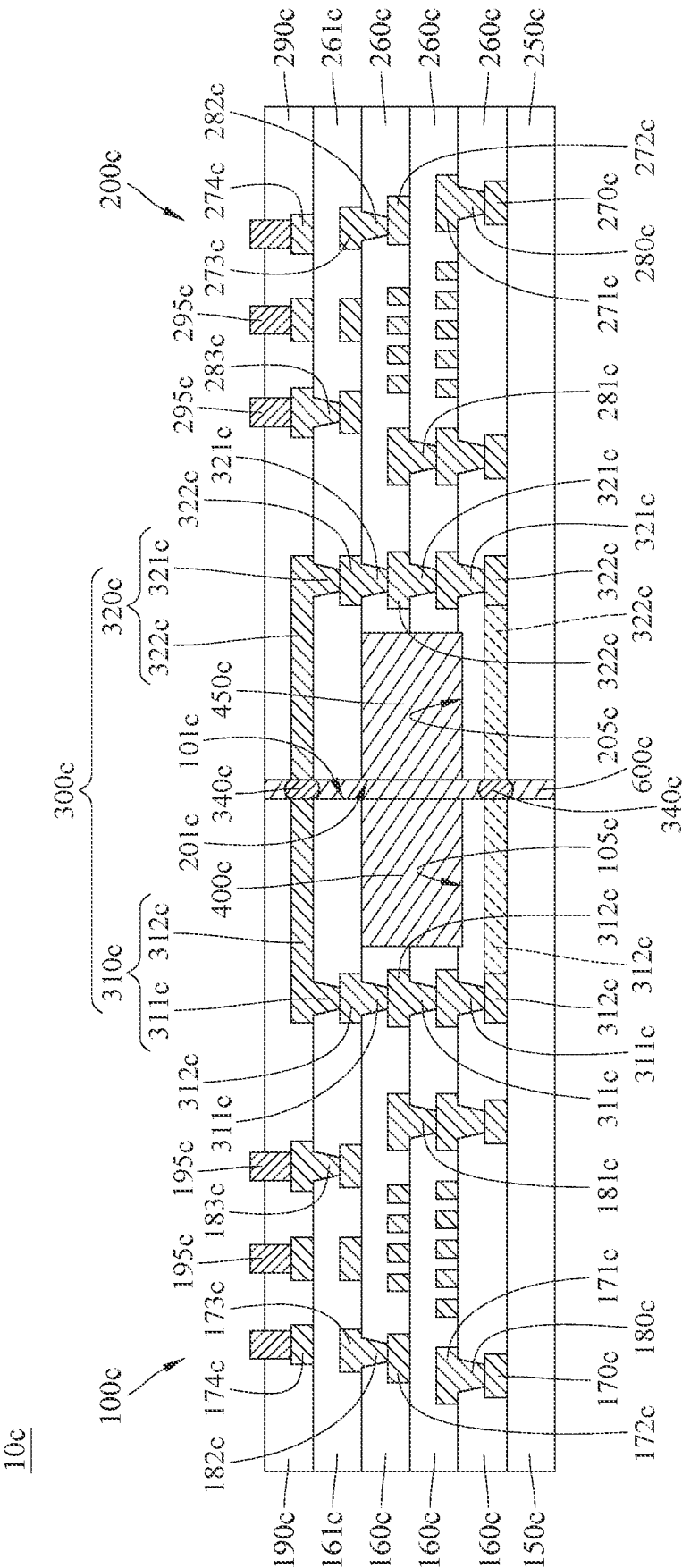


FIG. 20

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CIRCUIT BOARD STRUCTURE AND MANUFACTURING METHOD THEREOF

CROSS-REFERENCE TO RELATED APPLICATIONS

This non-provisional application claims priority under 35 U.S.C. § 119(a) on Patent Application No(s). 112118639 filed in Taiwan, R.O.C. on May 19, 2023, the entire contents of which are hereby incorporated by reference.

TECHNICAL FIELD

The disclosure relates to a circuit board structure and a manufacturing method thereof, more particularly to a circuit board structure including at least one conductive coil and at least one magnetic body and a manufacturing method thereof.

BACKGROUND

An inductor, a passive component in electronics, has multiple functions of, for example, filtering noise, suppressing momentary current, preventing interference caused by electromagnetic wave, shielding electromagnetic radiation, reducing electromagnetic interference and converting power. Thus, there is usually at least one inductor disposed in a circuit board.

In general, in order to configure an inductor, an entire of a conductive coil is formed in a single circuit board. However, the entire of the conductive coil occupies significant amount of space in the circuit board, which is unfavorable for the space utilization and the miniaturization of the circuit board.

SUMMARY

The disclosure provides a circuit board structure and a manufacturing method thereof to improve the space utilization of the circuit board and facilitate the miniaturization of the circuit board.

One embodiment of this disclosure provides a circuit board structure including a first circuit board, a second circuit board, a conductive coil, a first magnetic body and a molding compound. The first circuit board has a first side surface and a first cavity located on the first side surface. The second circuit board has a second side surface facing the first side surface and being spaced apart from the first side surface. The conductive coil is in a spiral shape and includes a first coil pattern and a second coil pattern. The first coil pattern is disposed in the first circuit board. The second coil pattern is disposed in the second circuit board. The first coil pattern is electrically connected to the second coil pattern. The first magnetic body is filled in the first cavity of the first circuit board. The conductive coil surrounds at least a part of the first magnetic body. The molding compound is filled in a gap between the first side surface and the second side surface.

In an embodiment of the disclosure, the circuit board structure further includes a second magnetic body. The second circuit board further has a second cavity. The second cavity is located on the second side surface. The second magnetic body is filled in the second cavity. The conductive coil surrounds at least a part of the second magnetic body.

In an embodiment of the disclosure, the circuit board structure further includes a first chip and a second chip. The

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first chip is disposed on the first circuit board. The second chip is disposed on the second circuit board.

In an embodiment of the disclosure, a size of the first circuit board is larger than a size of the second circuit board.

In an embodiment of the disclosure, the molding compound is magnetic.

In an embodiment of the disclosure, an inductance formed by the conductive coil and the first magnetic body is proportional to an effective cross-sectional area of the first magnetic body.

Another embodiment of this disclosure provides a manufacturing method of a circuit board structure including forming a first coil pattern of a conductive coil and a first magnetic body in a first circuit board; forming a second coil pattern of the conductive coil in a second circuit board; electrically connecting the first coil pattern to the second coil pattern; and filling a molding compound in a gap between the first side surface and a second side surface of the second circuit board. The conductive coil surrounds at least a part of the first magnetic body.

In an embodiment of the disclosure, the forming the first coil pattern and the first magnetic body in the first circuit board comprises forming an inner circuit pattern in a circuit substrate; forming a first cavity in the circuit substrate; filling a magnetic material in the first cavity; forming an outer circuit pattern in the circuit substrate; and cutting a part of the circuit substrate to form the first circuit board and allow the first cavity to be located on a first side surface of the first circuit board, cutting a part of the inner circuit pattern and a part of the outer circuit pattern to form the first coil pattern, and cutting a part of the magnetic material to form the first magnetic body.

In an embodiment of the disclosure, the manufacturing method of the circuit board structure further includes forming a second magnetic body in the second circuit board before electrically connecting the first coil pattern to the second coil pattern. The conductive coil surrounds at least a part of the second magnetic body.

In an embodiment of the disclosure, the manufacturing method of the circuit board structure further includes disposing a first chip on the first circuit board and disposing a second chip on the second circuit board.

According to the circuit board structure and the manufacturing method thereof disclosed by above embodiments, the conductive coil is in a spiral shape and includes the first coil pattern disposed in the first circuit board and the second coil pattern disposed in the second circuit board. That is, the present disclosure respectively forms multiple coil patterns in multiple circuit boards to configure the conductive coil, thereby preventing the conductive coil from occupying significant amount of space in a single circuit board. Accordingly, the space utilization of each of the first circuit board and the second circuit board is improved, and the miniaturization of each of the first circuit board and the second circuit board is facilitated.

Further, the conductive coil surrounds at least a part of the first magnetic body. Thus, the first magnetic body can enhance the inductance effect generated by the conductive coil. Also, the first magnetic body is filled in the first cavity located on the first side surface. Thus, the first magnetic body can have large filling volume and large effective cross-sectional area, and the conductive coil can have more turns, thereby further enhancing the inductance effect generated by the conductive coil.

BRIEF DESCRIPTION OF THE DRAWINGS

The present disclosure will become better understood from the detailed description given herein below and the

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accompanying drawings which are given by way of illustration only and thus are not intending to limit the present disclosure and wherein:

FIGS. 1 to 7 are side cross-sectional views of a manufacturing method of a circuit board structure according to a first embodiment of the disclosure;

FIG. 8 is a top view of the circuit board structure in FIG. 7;

FIG. 9 is a perspective view of a conductive coil of the circuit board structure in FIG. 7;

FIG. 10 is a side cross-sectional view of a circuit board structure according to a second embodiment of the disclosure;

FIG. 11 is a side cross-sectional view of a circuit board structure according to a third embodiment of the disclosure; and

FIGS. 12 to 20 are side cross-sectional views of a manufacturing method of a circuit board structure according to a fourth embodiment of the disclosure.

DETAILED DESCRIPTION

In the following detailed description, for purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding of the disclosed embodiments. It will be apparent, however, that one or more embodiments may be practiced without these specific details. In other instances, well-known structures and devices are schematically shown in order to simplify the drawing.

Please refer to FIGS. 1 to 7. FIGS. 1 to 7 are side cross-sectional views of a manufacturing method of a circuit board structure 10 according to a first embodiment of the disclosure. The manufacture method of the circuit board structure 10 according to this embodiment may include the following steps.

Please refer to FIG. 1, an inner circuit pattern 30 is formed in the circuit substrate 20. The circuit substrate 20 includes a substrate 150, dielectric layers 160, wiring layers 170, 171 and 172 and conductive blind vias 180 and 181. The dielectric layers 160 are disposed on the substrate 150. The wiring layers 170, 171 and 172 are disposed on the dielectric layers 160 or the substrate 150, respectively. The conductive blind vias 180 and 181 are located in the dielectric layers 160, respectively. The inner circuit pattern 30 includes vertical portions 311 and horizontal portions 312. The vertical portions 311 of the inner circuit pattern 30 are formed together with the conductive blind vias 180 and 181 in the dielectric layers 160. The horizontal portions 312 of the inner circuit pattern 30 are formed together with the wiring layers 170, 171 and 172 on the dielectric layers 160 or the substrate 150.

Please refer to FIG. 2, a first cavity 105 is formed in the circuit substrate 20. The first cavity 105 is located on a top surface 21 of the circuit substrate 20. Further, the first cavity 105 exposes a part of one of the horizontal portions 312 of the inner circuit pattern 30. In this embodiment, the first cavity 105 passes through, for example, three dielectric layers 160, but the disclosure is not limited thereto. In other embodiments, the first cavity may pass through one, two or at least four dielectric layers.

Please refer to FIG. 3, a magnetic material 40 is filled in the first cavity 105.

Please refer to FIG. 4, an outer circuit pattern 31 is formed in the circuit substrate 20, bonding pads 173 are formed on the dielectric layers 160, and a conductive blind via 182 is formed in the dielectric layer 160. The outer circuit pattern

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31 includes vertical portions 311 and horizontal portions 312. The vertical portions 311 of the outer circuit pattern 31 are formed together with the conductive blind via 182 in the dielectric layer 160. The horizontal portions 312 of the outer circuit pattern 31 are formed together with the bonding pads 173 on the dielectric layers 160.

Please refer to FIG. 5, an insulating material layer 190 is formed on the dielectric layers 160, and conductive bumps 195 are formed on the bonding pads 173.

Please refer to FIGS. 5 and 6, a part of the circuit substrate 20 is cut to form a first circuit board 100 and to allow the first cavity 105 to be located on a first side surface 101 of the first circuit board 100. Also, a part of the inner circuit pattern 30 and a part of the outer circuit pattern 31 are cut to form a first coil pattern 310, and a part of the magnetic material 40 is cut to form a first magnetic body 400. A step of forming the first coil pattern 310 and the first magnetic body 400 in the first circuit board 100 is completed so far. The first coil pattern 310 includes the vertical portions 311 and the horizontal portions 312.

Please refer to FIG. 7, a second coil pattern 320 of a conductive coil 300 is formed in a second circuit board 200, and a second magnetic body 450 is formed in the second circuit board 200. The step of forming the second coil pattern 320 of the conductive coil 300 in the second circuit board 200 and the step of forming the second magnetic body 450 in the second circuit board 200 may be simultaneously performed in a manner similar to the step of forming the first coil pattern 310 and the first magnetic body 400 in the first circuit board 100.

The second circuit board 200 has a second side surface 201 and a second cavity 205. The second cavity 205 is located on the second side surface 201. The second side surface 201 faces the first side surface 101, and is spaced apart from the first side surface 101. The second circuit board 200 includes a substrate 250, dielectric layers 260, wiring layers 270, 271 and 272, bonding pads 273, conductive blind vias 280, 281 and 282, an insulating material layer 290 and conductive bumps 295.

The dielectric layers 260 are disposed on the substrate 250. The wiring layers 270, 271 and 272 and the bonding pads 273 are disposed on the dielectric layers 260, respectively. The conductive blind vias 280, 281 and 282 are located in the dielectric layers 260, respectively. The insulating material layer 290 is located on the dielectric layers 260. The conductive bumps 295 are located on the bonding pads 273, respectively. The second coil pattern 320 includes vertical portions 321 and horizontal portions 322. The vertical portions 321 are formed together with the conductive blind vias 280, 281 and 282 in the dielectric layers 260, respectively. The horizontal portions 322 are formed together with the wiring layers 270, 271 and 272 and the bonding pads 273 on the dielectric layers 260 or the substrate 250. The second magnetic body 450 is filled in the second cavity 205.

In addition, the first coil pattern 310 is electrically connected to the second coil pattern 320 via, for example, solder pads 330.

In addition, a molding compound 600 is filled in a gap between the first side surface 101 and the second side surface 201. Manufacturing of the circuit board structure 10 is completed so far.

In this embodiment, the molding compound 600 is, for example, magnetic, but the disclosure is not limited thereto. In other embodiments, the molding compound may not be magnetic.

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Note that in this embodiment, an insulating frame part (i.e., the part of the circuit substrate **20**, the part of the inner circuit pattern **30**, the part of the outer circuit pattern **31** and the part of the magnetic material **40** that are cut in FIG. **6**) excluding the first circuit board **100** is cut before performing the packaging process using the molding compound **600** in FIG. **7**. Thus, the available space for the wirings in the resulting first circuit board **100** is prevented from being reduced by the cutting step in FIG. **6**.

Note that the disclosure is not limited by the order of the steps shown in FIGS. **1** to **7**. For example, in this embodiment, the first coil pattern **310** and the first magnetic body **400** are formed in the first circuit board **100** before the second coil pattern and the second magnetic body are formed in the second circuit board, but the disclosure is not limited thereto. In other embodiments, in the same step, the first coil pattern and the first magnetic body may be formed in the first circuit board while the second coil pattern and the second magnetic body are formed in the second circuit board.

Please refer to FIGS. **8** and **9**. FIG. **8** is a top view of the circuit board structure **10** in FIG. **7**. FIG. **9** is a perspective view of the conductive coil **300** of the circuit board structure **10** in FIG. **7**. In this embodiment, the conductive coil **300** is in a spiral shape and includes the first coil pattern **310**, the second coil pattern **320** and solder pads **330**. Furthermore, the conductive coil **300** surrounds at least a part of the first magnetic body **400** and at least a part of the second magnetic body **450**. In addition, in order to illustrate the spiral shape of the conductive coil **300**, FIG. **9** shows the conductive coil **300** with simplified shape. In practical, the vertical portions **311** of the first coil pattern **310** and the vertical portions **321** of the second coil pattern **320** in FIG. **9** may be similar to the conductive blind vias **180**, **181**, **182**, **280**, **281** and **282** in shape. Also, the horizontal portions **312** of the first coil pattern **310** and the horizontal portions **322** of the second coil pattern **320** in FIG. **9** may be similar to the wiring layers **170**, **171**, **172**, **270**, **271** and **272** or the bonding pads **173** and **273** in shape. As shown in FIG. **9**, the conductive coil **300** has, for example, two and a half turns, but the disclosure is not limited thereto. In other embodiments, the conductive coil may merely have a single turn.

In addition, the disclosure is not limited by the arranging direction of the turns of the conductive coil **300**. In other embodiments, as long as the conductive coil surrounds at least a part of the first molding compound, the turns of the conductive coil are allowed to be arranged along an arbitrary direction.

In this disclosure, the first coil pattern **310** and the second coil pattern **320** are respectively formed in the first circuit board **100** and the second circuit board **200** to configure the conductive coil **300**. Thus, the space of the first circuit board **100** or the second circuit board **200** occupied by the conductive coil **300** is reduced. In this way, the space utilization of each of the first circuit board **100** and the second circuit board **200** is improved, and the miniaturization of each of the first circuit board **100** and the second circuit board **200** is facilitated.

Further, the conductive coil **300** surrounds at least a part of the first magnetic body **400**. Thus, the first magnetic body **400** can enhance the inductance effect generated by the conductive coil **300**. Also, the first magnetic body **400** is filled in the first cavity **105** located on the first side surface **101**. Thus, the first magnetic body **400** can have large filling volume (i.e., the volume of a part of the first magnetic body **400** that is filled in the first cavity **105**) and large effective cross-sectional area, and the conductive coil can have more

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turns, thereby further enhancing the inductance effect generated by the conductive coil **300**.

Additionally, the inductance effect generated by the conductive coil **300**, the first magnetic body **400** and the second magnetic body **450** depends on the filling volumes of the first magnetic body **400** and the second magnetic body **450** and the turns of the conductive coil **300**. Specifically, if the filling volumes of the first magnetic body **400** and the second magnetic body **450** or the turns of the conductive coil **300** increases, the generated inductance effect will be increased; in contrast, if the filling volumes of the first magnetic body **400** and the second magnetic body **450** or the turns of the conductive coil **300** decreases, the generated inductance effect will be decreased.

In this embodiment, the inductance effect generated by the conductive coil **300** and the first magnetic body **400** is proportional to the effective cross-sectional area of the first magnetic body **400**. In detail, the inductance formed by the conductive coil **300** and the first magnetic body **400** can be obtained by, for example, the following formula:

$$L = \frac{N^2 \mu A_e}{l_e}$$

In the above formula, L denotes the inductance formed by the conductive coil **300** and the first magnetic body **400**; N denotes a number of the turn(s) of the conductive coil **300**; μ denotes permeability; A_e denotes an effective cross-sectional area of the first magnetic body **400**, where the effective cross-sectional area may be regarded as the contact area or the area of the orthogonal projection between the conductive coil **300** and the first magnetic body **400**, and in this embodiment, the effective cross-sectional area is, for example, the contact area between the conductive coil **300** and the first magnetic body **400**; l_e denotes an effective magnetic path length of the first magnetic body **400**. According to the above formula, if the filling area of the first magnetic body **400** increases, not only the effective cross-sectional area A_e is increased, but also the conductive coil **300** may have more turns, thereby enhancing the inductance formed by the conductive coil **300** and the first magnetic body **400**. Note that the inductance is formed by the conductive coil **300** and the second magnetic body **450** based on a similar principle, and thus the repeated descriptions are omitted.

In this embodiment, the filling volume of the first magnetic body **400** is the same as the filling volume of the second magnetic body **450**, but the disclosure is not limited thereto. In other embodiments, the filling volume of the first magnetic body may be different from the filling volume of the second magnetic body. Correspondingly, the size of the first cavity **105** and the size of the second cavity **205** may be designed based on actual requirements of the filling volume of the first magnetic body **400** and the filling volume of the second magnetic body **450**.

In this embodiment, as shown in FIG. **7**, a top surface **401** and a bottom surface **402** of the first magnetic body **400** are flush with a top surface **451** and a bottom surface **452** of the second magnetic body **450**, respectively, but the disclosure is not limited thereto. In other embodiments, the top surface and the bottom surface of the first magnetic body may be misaligned with (i.e., may not be flushed with) the top surface and the bottom surface of the second magnetic body, respectively.

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In this embodiment, the molding compound **600** is filled in the gap between the first side surface **101** and the second side surface **201**. That is, as shown in FIG. 7, a top surface **102** or a top surface **202** of one of the first circuit board **100** and the second circuit board **200** is not located lower than a bottom surface **103** or a bottom surface **203** of the other of the first circuit board **100** and the second circuit board **200**. Similarly, the bottom surface **103** or the bottom surface **203** of one of the first circuit board **100** and the second circuit board **200** is not located higher than the top surface **102** or the top surface **202** of the other of the first circuit board **100** and the second circuit board **200**. With such configuration, at least a part of the first side surface **101** of the first circuit board **100** faces the second side surface **201** of the second circuit board **200**, and thus the gap for the molding compound **600** to be filled therein is formed.

Note that in other embodiments, the first circuit board and the second circuit board may be disposed on a third circuit board to configure a circuit board structure including three circuit boards. Alternatively, in other embodiments, the first circuit board and the second circuit board may be disposed on a third circuit board, and a fourth circuit board may be disposed on the first circuit board and the second circuit board to configure a circuit board structure including four circuit boards.

In this embodiment, the first magnetic body **400** and the second magnetic body **450** are respectively filled in the first circuit board **100** and the second circuit board **200**, and the first circuit board **100** and the second circuit board **200** have the same size. However, the disclosure is not limited thereto.

Other embodiments are described below for illustrative purposes. It is to be noted that the following embodiments use the reference numerals and a part of the contents of the above embodiments, the same reference numerals are used to denote the same or similar elements, and the description of the same technical contents is omitted. For the description of the omitted part, reference may be made to the above embodiments, and details are not described in the following embodiments.

Please refer to FIG. 10. FIG. 10 is a side cross-sectional view of a circuit board structure **10a** according to a second embodiment of the disclosure. Hereinafter, only the different between the circuit board structure **10a** of this embodiment and the circuit board structure **10** of the first embodiment will be described, and other similar or identical descriptions are omitted. In this embodiment, a second circuit board **200a** of the circuit board structure **10a** does not have the second cavity **205** in the first embodiment, and no magnetic body (e.g. second magnetic body **450** in the first embodiment) is filled in the second circuit board **200a**. Moreover, a size of the first circuit board **100** is larger than a size of the second circuit board **200a**. However, the disclosure is not limited thereto.

Please refer to FIG. 11. FIG. 11 is a side cross-sectional view of a circuit board structure **10b** according to a third embodiment of the disclosure. Hereinafter, only the different between the circuit board structure **10b** of this embodiment and the circuit board structure **10** of the first embodiment will be described, and other similar or identical descriptions are omitted. Comparing to the circuit board structure **10** of the first embodiment, the circuit board structure **10b** of this embodiment further includes a first chip **650b**, a second chip **660b**, a second molding compound **670b** and a third molding compound **680b**. The first chip **650b** is electrically connected to the conductive bumps **195** of the first circuit board **100**. The second chip **660b** is electrically connected to the conductive bumps **295** of the second circuit board **200**. The

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second molding compound **670b** is filled in a gap between the first chip **650b** and the insulating material layer **190**, and in a gap between the second chip **660b** and the insulating material layer **290**. The third molding compound **680b** is disposed on the first chip **650b**, the second chip **660b** and the insulating material layers **190** and **290**.

Comparing to the manufacturing method of the circuit board structure **10** of the first embodiment, a manufacturing method of the circuit board structure **10b** may further include a step of disposing the first chip **650b** on the first circuit board **100** and disposing the second chip **660b** on the second circuit board **200**.

In this disclosure, the first magnetic body and the second magnetic body are not limited to respectively be in contact with the first coil pattern and the second coil pattern, and the first coil pattern is not limited to be electrically connected to the second coil pattern via the solder pads. Please refer to FIGS. 12 to 20. FIGS. 12 to 20 are side cross-sectional views of a manufacturing method of a circuit board structure **10c** according to a fourth embodiment of the disclosure.

Please refer to FIG. 12, an inner circuit pattern **30c** is formed in a circuit substrate **20c**. The circuit substrate **20c** includes a substrate **150c**, dielectric layers **160c**, wiring layers **170c**, **171c** and **172c** and conductive blind vias **180c** and **181c**. The dielectric layers **160c** are disposed on the substrate **150c**. The wiring layers **170c**, **171c** and **172c** are disposed on the dielectric layers **160c** or the substrate **150c**, respectively. The conductive blind vias **180c** and **181c** are located in the dielectric layers **160c**, respectively. The inner circuit pattern **30c** includes vertical portions **311c** and horizontal portions **312c**. The vertical portions **311c** of the inner circuit pattern **30c** are formed together with the conductive blind vias **180c** and **181c** in the dielectric layers **160c**. The horizontal portions **312c** of the inner circuit pattern **30c** are formed together with the wiring layers **170c**, **171c** and **172c** on the dielectric layers **160c** or the substrate **150c**.

Please refer to FIG. 13, a first cavity **105c** is formed in the circuit substrate **20c**. The first cavity **105c** is located on a top surface **21c** of the circuit substrate **20c**, and the horizontal portions **312c** of the inner circuit pattern **30c** are spaced apart from the first cavity **105c**. That is, the first cavity **105c** does not expose the horizontal portions **312c** of the inner circuit pattern **30c**.

Please refer to FIG. 14, a magnetic material **40c** is filled in the first cavity **105c**.

Please refer to FIG. 15, a build-up wiring layer **173c** is formed on the dielectric layers **160c**, a conductive blind via **182c** is formed in the dielectric layer **160c**, and a first outer circuit pattern **31c** is formed in the circuit substrate **20c**. The first outer circuit pattern **31c** includes vertical portions **311c** and horizontal portions **312c**. The vertical portions **311c** of the first outer circuit pattern **31c** are formed together with the conductive blind via **182c** in the dielectric layer **160c**. The horizontal portions **312c** of the first outer circuit pattern **31c** are formed together with the build-up wiring layer **173c** on the dielectric layers **160c**.

Please refer to FIG. 16, a build-up dielectric layer **161c** is formed on the dielectric layers **160c**, the build-up wiring layer **173c** and the horizontal portions **312c** of the first outer circuit pattern **31c**.

Please refer to FIG. 17, a second outer circuit pattern **32c** is formed in the build-up dielectric layer **161c**, bonding pads **174c** are formed on the build-up dielectric layer **161c**, and a conductive blind via **183c** is formed in the build-up dielectric layer **161c**. The second outer circuit pattern **32c** includes vertical portions **311c** and horizontal portions **312c**. The vertical portions **311c** of the second outer circuit pattern

32c are formed together with the conductive blind via 183c in the build-up dielectric layer 161c. The horizontal portions 312c of the second outer circuit pattern 32c are formed together with the bonding pads 174c on the build-up dielectric layer 161c.

Please refer to FIG. 18, an insulating material layer 190c is formed on the build-up dielectric layer 161c, and conductive bumps 195c are formed on the bonding pads 174c.

Please refer to FIGS. 18 and 19, a part of the circuit substrate 20c is cut to form a first circuit board 100c and to allow the first cavity 105c to be located on a first side surface 101c of the first circuit board 100c. Also, a part of the inner circuit pattern 30c, and a part of the second outer circuit pattern 32c are cut to form a first coil pattern 310c, and a part of the magnetic material 40c is cut to form a first magnetic body 400c. A step of forming the first coil pattern 310c and the first magnetic body 400c in the first circuit board 100c is completed so far. The first coil pattern 310c includes vertical portions 311c and horizontal portions 312c. Further, in this embodiment, the first magnetic body 400c is spaced apart from the first coil pattern 310c. Thus, in this embodiment, the effective cross-sectional area is regarded as, for example, the area of the orthogonal projection between the conductive coil 300c and the first magnetic body 400c.

Please refer to FIG. 20, a second coil pattern 320c of the conductive coil 300c is formed in a second circuit board 200c, and a second magnetic body 450c is formed in the second circuit board 200c.

The second circuit board 200c has a second side surface 201c and a second cavity 205c. The second circuit board 200c includes a substrate 250c, dielectric layers 260c, a build-up dielectric layer 261c, wiring layers 270c, 271c and 272c, a build-up wiring layer 273c, bonding pads 274c, conductive blind vias 280c, 281c, 282c and 283c, an insulating material layer 290c and conductive bumps 295c.

The second circuit board 200c and the first circuit board 100c are similar in structure, and thus the repeated descriptions are omitted. In addition, the connection relationships between the second circuit board 200c, the second coil pattern 320c and the second magnetic body 450c are similar to that between the first circuit board 100c, the first coil pattern 310c and the first magnetic body 400c, and thus the repeated descriptions are omitted.

In addition, the first coil pattern 310c is electrically connected to the second coil pattern 320c via, for example, solder balls 340c. However, the disclosure is not limited thereto. In other embodiments, after a surface treatment is performed on the first side surface and the second side surface, the first coil pattern may be electrically connected to the second coil pattern via solder pads and solder balls.

Moreover, a molding compound 600c is filled in a gap between the first side surface 101c and the second side surface 201c. Manufacturing of the circuit board structure 10c is completed so far.

According to the circuit board structure and the manufacturing method thereof disclosed by above embodiments, the conductive coil is in a spiral shape and includes the first coil pattern disposed in the first circuit board and the second coil pattern disposed in the second circuit board. That is, the present disclosure respectively forms multiple coil patterns in multiple circuit boards to configure the conductive coil, thereby preventing the conductive coil from occupying significant amount of space in a single circuit board. Accordingly, the space utilization of each of the first circuit board and the second circuit board is improved, and the miniaturization of each of the first circuit board and the second circuit board is facilitated.

Further, the conductive coil surrounds at least a part of the first magnetic body. Thus, the first magnetic body can enhance the inductance effect generated by the conductive coil. Also, the first magnetic body is filled in the first cavity located on the first side surface. Thus, the first magnetic body can have large filling volume and large effective cross-sectional area, and the conductive coil can have more turns, thereby further enhancing the inductance effect generated by the conductive coil.

It will be apparent to those skilled in the art that various modifications and variations can be made to the present disclosure. It is intended that the specification and examples be considered as exemplary embodiments only, with a scope of the disclosure being indicated by the following claims and their equivalents.

What is claimed is:

1. A circuit board structure, comprising:

a first circuit board, having a first side surface and a first cavity located on the first side surface;

a second circuit board, having a second side surface facing the first side surface and being spaced apart from the first side surface;

a conductive coil, in a spiral shape and comprising a first coil pattern and a second coil pattern, wherein the first coil pattern is disposed in the first circuit board, the second coil pattern is disposed in the second circuit board, and the first coil pattern is electrically connected to the second coil pattern;

a first magnetic body, filled in the first cavity of the first circuit board, the conductive coil surrounding at least a part of the first magnetic body; and

a molding compound, filled in a gap between the first side surface and the second side surface,

wherein the first side surface and the second side surface are parallel to a height direction of the first magnetic body.

2. The circuit board structure according to claim 1, further comprising a second magnetic body, wherein the second circuit board further has a second cavity, the second cavity is located on the second side surface, the second magnetic body is filled in the second cavity, and the conductive coil surrounds at least a part of the second magnetic body.

3. The circuit board structure according to claim 2, further comprising a first chip and a second chip, wherein the first chip is disposed on the first circuit board, and the second chip is disposed on the second circuit board.

4. The circuit board structure according to claim 1, wherein a size of the first circuit board is larger than a size of the second circuit board.

5. The circuit board structure according to claim 1, wherein the molding compound is magnetic.

6. The circuit board structure according to claim 1, wherein an inductance formed by the conductive coil and the first magnetic body is proportional to an effective cross-sectional area of the first magnetic body.

7. A manufacturing method of a circuit board structure, comprising:

forming a first coil pattern of a conductive coil and a first magnetic body in a first circuit board;

forming a second coil pattern of the conductive coil in a second circuit board;

electrically connecting the first coil pattern to the second coil pattern; and

filling a molding compound in a gap between a first side surface of the first circuit board and a second side surface of the second circuit board, wherein the first

side surface and the second side surface are parallel to a height direction of the first magnetic body; and wherein, the conductive coil surrounds at least a part of the first magnetic body.

8. The manufacturing method of the circuit board structure according to claim 7, wherein the forming the first coil pattern and the first magnetic body in the first circuit board comprises:

forming an inner circuit pattern in a circuit substrate;
forming a first cavity in the circuit substrate;
filling a magnetic material in the first cavity;
forming an outer circuit pattern in the circuit substrate;
and

cutting a part of the circuit substrate to form the first circuit board and allow the first cavity to be located on a first side surface of the first circuit board, cutting a part of the inner circuit pattern and a part of the outer circuit pattern to form the first coil pattern, and cutting a part of the magnetic material to form the first magnetic body.

9. The manufacturing method of the circuit board structure according to claim 7, further comprising forming a second magnetic body in the second circuit board before electrically connecting the first coil pattern to the second coil pattern, wherein the conductive coil surrounds at least a part of the second magnetic body.

10. The manufacturing method of the circuit board structure according to claim 7, further comprising disposing a first chip on the first circuit board and disposing a second chip on the second circuit board.

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